

Fictional sample created for software testing. All names, entities, numbers, and dates are invented.

EXHIBIT A

STATEMENT OF PROPOSED ENDEAVOR

Ilona Marta Brasch, Ph.D.
4177 Quarry Bend Drive, Apt. 12B, Sparks, NV 89436
In support of Form I-140, EB-2 National Interest Waiver
Dated: October 28, 2025

1. WHO I AM

I am a chemical engineer. For eleven years I have worked on one narrow problem: how to pull a single ion out of a hot, dirty, chemically crowded water stream without pulling out everything else with it. I did my doctoral work on layered manganese-oxide sorbents in Tartu, spent three years as a postdoctoral researcher at the Cascadia Institute of Technology in Portland, Oregon, and in 2023 I founded Halide Loop Technologies, Inc. in Sparks, Nevada, where I am Chief Technology Officer.

2. WHAT I PROPOSE TO DO

I will continue to develop, field-qualify, and place into commercial operation the HL-2 sorbent membrane module, a direct lithium extraction unit that recovers battery-grade lithium chloride from geothermal brine at the wellhead.

Stated concretely, over the next four years I will:

1. complete the 500-hour endurance qualification of the HL-2 module on the brine chemistry of the Pyramid Ridge field, Washoe County, Nevada;
2. design and commission a 12-module skid rated at 40 tonnes of lithium carbonate equivalent per year, sited at an operating geothermal plant;
3. publish the sorbent regeneration protocol and the fouling data openly, so that operators other than my own company can evaluate the method against their brines;
4. qualify a second sorbent formulation for the higher-calcium brines of the Black Rock corridor; and
5. train and lead the process engineering team that carries the design into routine operation.

The work will be carried out through Halide Loop Technologies, Inc., 2290 Vassar Industrial Way, Suite 140, Sparks, NV 89431, at its own laboratory and at host geothermal sites under site access agreements.

3. WHY THE WORK MATTERS

Lithium is a critical mineral, and the United States refines a very small share of the world's supply. The lithium that exists domestically in geothermal brine is largely stranded. Brine leaves a geothermal power plant at 130 to 190 degrees Celsius carrying lithium at 60 to 340 milligrams per litre, together with far larger quantities of sodium,

potassium, calcium and silica. The conventional route — evaporation ponds — needs years of residence time, enormous land area, and water losses that northern Nevada cannot afford. So the brine is reinjected, and the lithium goes back underground.

Direct extraction at the wellhead changes that arithmetic. It also changes it for someone other than me: the operators who would use it are power companies, not mining companies, and they do not have in-house separations chemistry. The endeavor is aimed at giving them a method they can adopt.

4. WHY THE IMPACT IS NATIONAL, NOT LOCAL

The HL-2 sorbent is not specific to one field. The same chemistry, with a different regeneration cycle, applies to:

- Basin and Range geothermal brines, across Nevada, Utah, and eastern California;
- oilfield produced water in the Permian and Williston basins, where lithium is present at lower concentration but volumes are very large; and
- spent cathode leachate in battery recycling, where the separation problem is the same one in a different matrix.

A domestic method that works on all three touches battery cell manufacturing, grid storage procurement, and the balance of critical mineral imports. It is not confined to the economics of one plant or one county.

5. WHAT HAS ALREADY BEEN ACHIEVED

This is not a proposal from a standing start. The record of the work to date:

Date	Milestone	Result
---	---	---
Mar 2021	HL-1 sorbent, bench columns, synthetic brine	Li/Na selectivity 340:1
Nov 2021	First real-brine trial, borrowed sample, Cascadia lab	Selectivity held at 291:1
Aug 2022	Regeneration cycle reduced from 14 h to 3.5 h	Published, *J. Selective Separations*
Jun 2023	Halide Loop Technologies, Inc. formed; HL-2 module designed	Module built in-house
Feb 2024	HL-2 skid installed on slipstream, Pyramid Ridge field	1,900 h continuous operation
Jan 2025	Sorbent loss per cycle measured across 400 cycles	0.11% per cycle
Jul 2025	Product purity from field brine	99.2% LiCl before polishing

The regeneration cycle described in the August 2022 paper is the subject of a pending patent application on which I am the sole inventor, and which is assigned to Halide Loop Technologies, Inc.

6. WHY THE TIMING MATTERS

The Nevada Division of Minerals has published the second round of geothermal brine mineral lease applications, with a closing date of March 31, 2027. An applicant must show an operating recovery method at the time of application, not a plan for one. The Pyramid Ridge slipstream is the only site at which the HL-2 module is installed, and the site

access agreement runs with the lease. If the 2027 round passes without an application, the demonstration site is not available again for the ten-year term of the leases awarded in that round.

7. WHAT I AM ASKING FOR

I have no employer to sponsor me. The company through which this work is done is one I founded and control, and the work depends on chemistry and operating knowledge I built over eleven years. I ask that the job offer requirement be waived so that I can carry this work through the field program and into commercial operation in the United States.

I declare under penalty of perjury that the foregoing is true and correct.

/s/ Ilona M. Brasch
Ilona Marta Brasch, Ph.D.
October 28, 2025